Teach & Repeat Rover Navigation

Project update | Recorded-data simulation

Tapisar Singh
Ayush Kaulgud
Irenov Joel

October 7, 2026

How we built the recorded-data test

- Recorded Freiburg University mapped data gave us the route and laser map.
- Python parses scans and plans paths. PyBullet drives a Husky rover at 120 Hz.

Recorded route

15 offset starts

Four planners

Bullet runs

All four methods share the same 15 starts, map, controller, and safety checks.

Recorded inputs; simulated repeat runs. FYI - ICP remains a separate test.

What was recorded and simulated

Recorded outdoors

Freiburg Campus, July 2004: Pioneer 2AT with a SICK laser.

The raw log contains 15,683 laser scans and odometry poses.

Scans 300–549 give a 29.59 m route; laser returns build the map.

Our repeat trials

Four planners × 15 common starts = 60 Bullet runs.

Offsets −1.2 to +1.2 m; headings −30°, 0°, +30°.

Completion, rejection, contacts, clearance, goal error, and time.

Sensor observations are from there, isplaced starts and repeat drives are simulated and logged odometry is a route reference, so not independent ground truth.

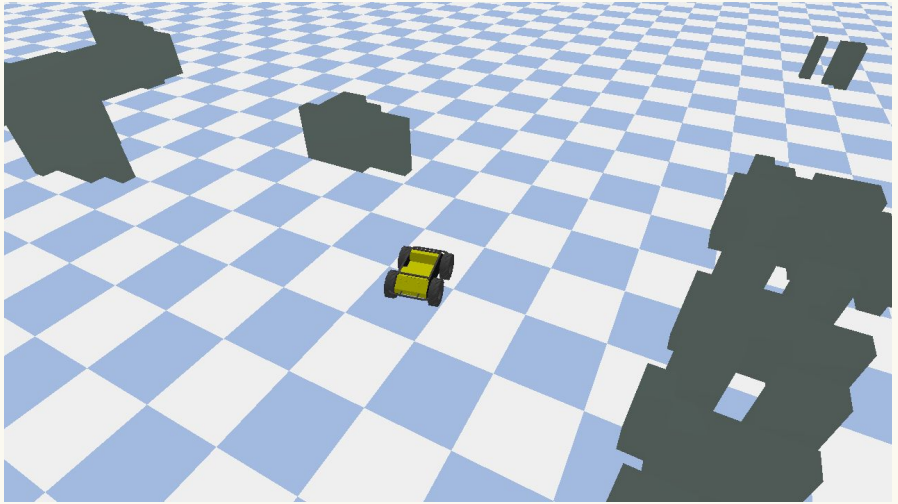
Recorded map, simulated run

RECORDED INPUT

29.59 m logged route
751 laser obstacle cells

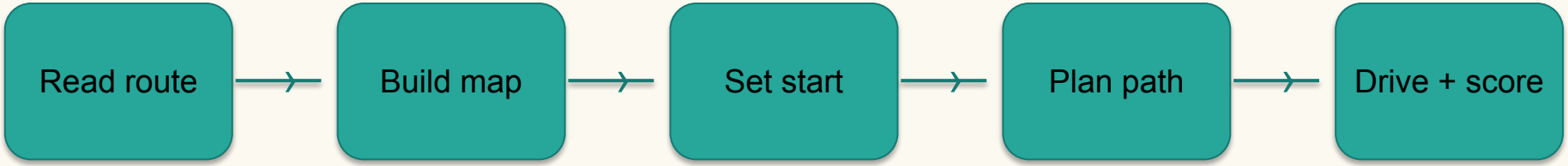
SIMULATED REPEAT

Four planners × 15 starts
PyBullet Husky surrogate
Shared map and controller
ICP tested separately



The obstacle mesh comes from the Pioneer 2AT's recorded laser scans.

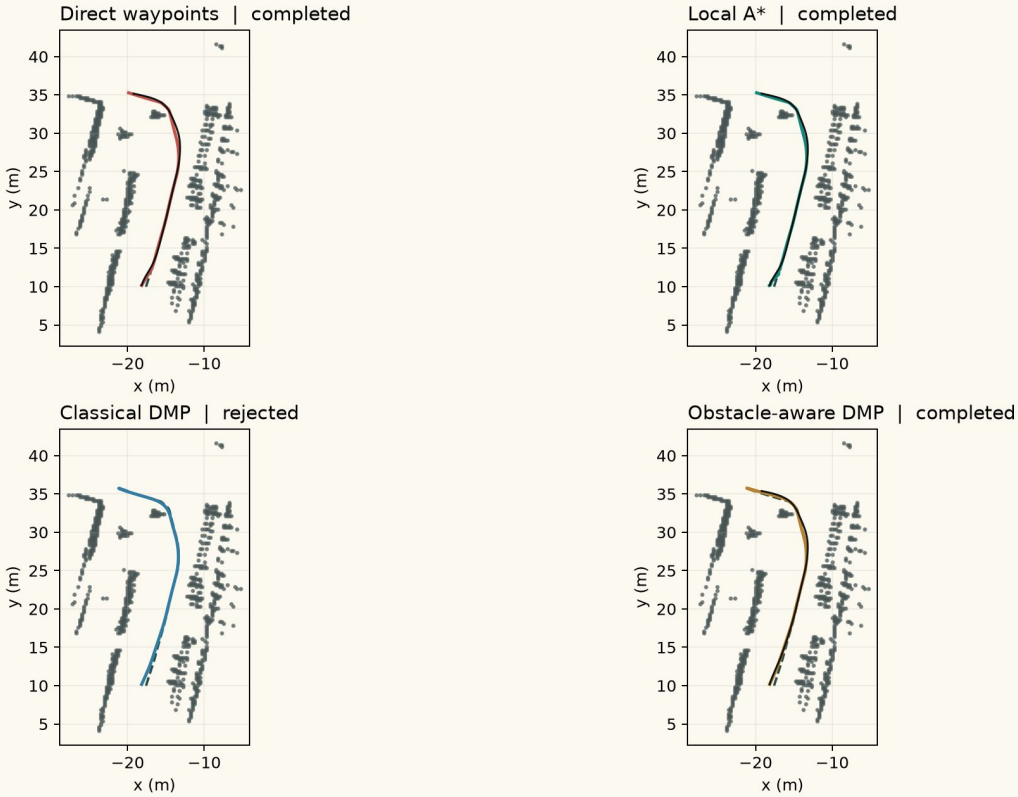
From recording to repeat run



Four candidate paths

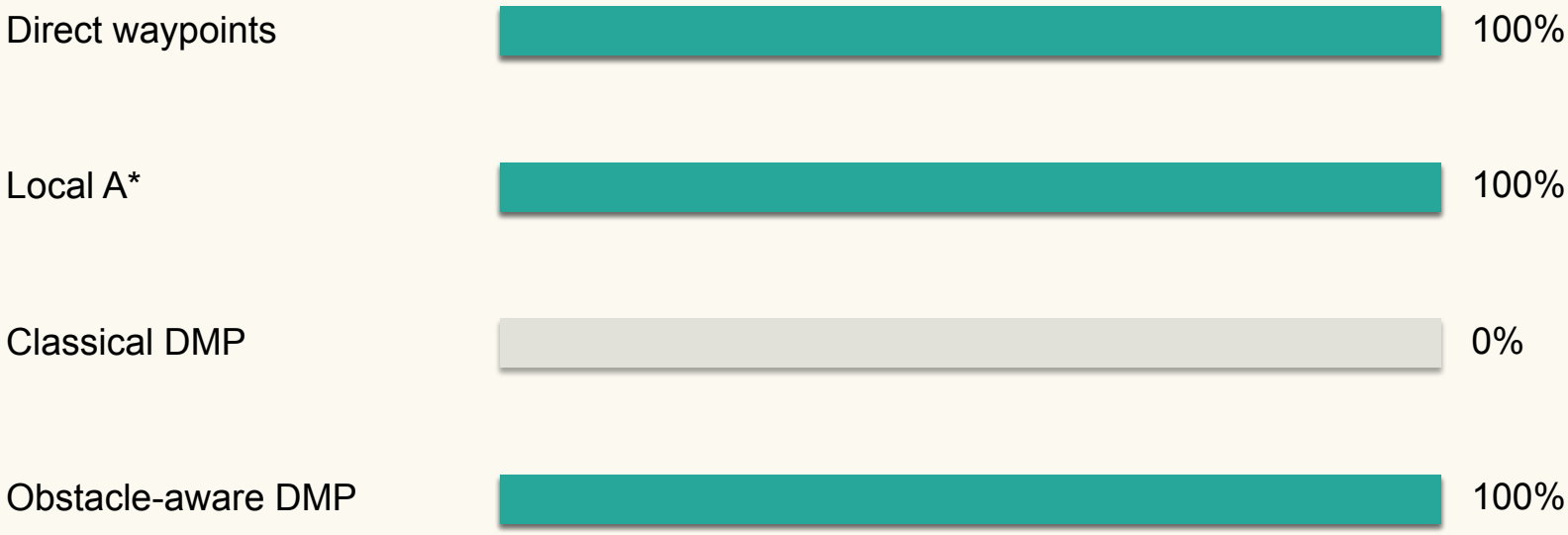
1. Direct waypoints
2. Local A* reconnection
3. Classical DMP
4. Obstacle-aware DMP

One offset start, four outcomes



Start: +0.6 m lateral, 0° heading. Gray points are recorded laser returns; black traces are Bullet runs.

Completion across 15 starts per method



Three methods completed all 15 starts; **classical DMP failed path clearance.**

0 obstacle contacts in 60 trials. Rejected paths were not driven.

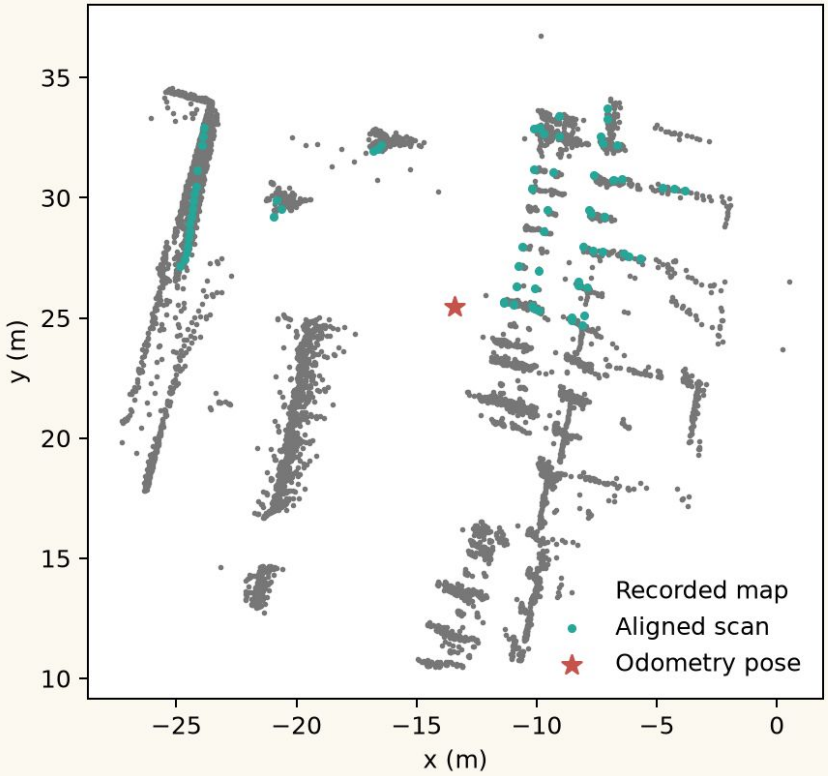
Other measurements from the same runs

Method	Rejected	Stopped	Min. clear.*	Goal err.	Plan time
Direct waypoints	0%	0%	0.287 m	0.646 m	0.12 ms
Local A*	0%	0%	0.287 m	0.646 m	1.59 ms
Classical DMP	100%	0%	N/A	N/A	7.73 ms
Obstacle-aware DMP	0%	0%	0.287 m	0.647 m	10.15 ms

No Bullet obstacle contacts. Goal error is the median among completed runs.

*Grid distance minus a 0.42 m body proxy; timing is planning only, on this PC.

ICP on a recorded laser scan



Held-out Freiburg scan #430

Initial pose offset 0.640 m

After ICP 0.044 m

Map residual 0.047 m

Compared with logged odometry, not ground truth. Navigation still uses exact Bullet pose.

What the results tell us

1. Direct, A*, and coupled DMP completed every start in this recorded segment.

2. Classical DMP failed the 0.15 m path margin; all 15 paths were rejected.

3. No Bullet obstacle contacts occurred with the safety screen enabled.

Next: test other routes and starts; use ICP during navigation; tune the rover model.

Method notes and sources

Trial design

15 starts = 5 lateral offsets × 3 headings. Route: 29.59 m. Laser map: 0.25 m. Path screen: 0.42 m body radius + 0.15 m margin. PyBullet Husky surrogate at 120 Hz.

Reproduce

python src/recorded_demo.py

GitHub: github.com/ayushkaulgud/teach-repeat-rover-navigation

Background references

Data: Stachniss & Grisetti, Freiburg Campus 2004 raw log (CC0 via MIT). Software: PyBullet. Code, CSV, plots, and slides: GitHub teach-repeat-rover-navigation.